Date: / /

Experiment No. 11

Aim: To prepare lyophobic sol (colloid) of ferric hydroxide.

Apparatus: Two beakers (250 mL), glass rod, funnel, Bunsen burner, filter paper, dropper etc.

Chemicals: Distilled water, 2 % ferric chloride solution prepared by dissolving 2 g of anhydrous pure $FeCl_3$ in 100 mL of distilled water.

Theory: Sols which cannot be formed simply by mixing substances such as metals, metal hydroxides or metal sulfides etc. with liquid dispersion medium (water) are called lyophobic sols (colloids).These sols are prepared by a separate method. Ferric hydroxide is a lyophobic sol.

Lyo means solvent and phobic means hating. Lyophobic sols are relatively less soluble than lyophilic sol. Ferric hydroxide sol is prepared by the hydrolysis of ferric chloride with boiling distilled water.

The reaction can be shown as-

$$FeCl_3(aq) + 3H_2O(l) \xrightarrow{\text{boil}} Fe(OH)_3(s) + 3HCl(aq)$$

Procedure:

1. Take 100 mL of distilled water in a clean 250 mL beaker and heat it to boil.
2. Add 2 % of ferric chloride solution dropwise with the help of dropper to the boiling water.
3. Continue heating until deep red or brown solution of ferric hydroxide is obtained.

Note: Add some water to replace the water lost by evaporation during boiling at regular intervals.

4. Keep the containt of the beaker undisturbed for some time at room temperature.
5. Filter the solution to get sol of $Fe(OH)_3$.

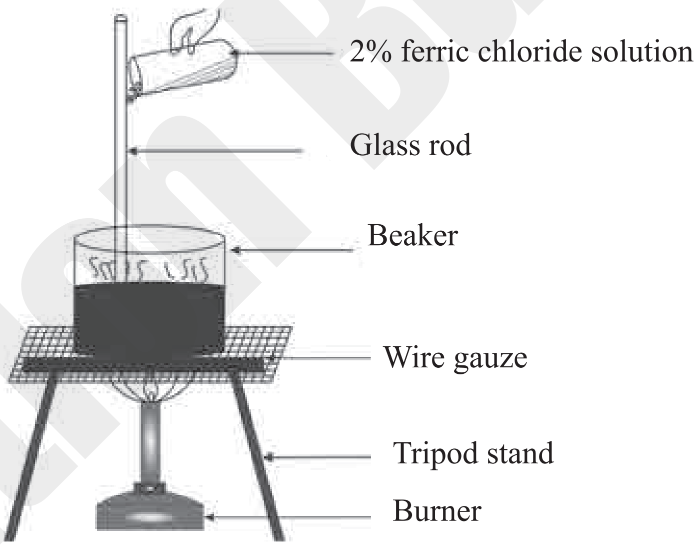

Fig. 11.1 Pouring of 2 % $FeCl_3$ solution in boiling water.

Observation:

Dark red brown sol of ferric hydroxide is formed which cannot be filtered easily through filter paper.

Conclusion:

A dark reddish-brown lyophobic sol of ferric hydroxide [Fe(OH)] is successfully prepared by the hydrolysis of ferric chloride (FeCl) with boiling water.

Remark and sign of teacher: ....

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. In lyophobic colloid the affinity between the particles of dispersed phase and dispersion medium is ....
   a. strong b. high c. low d. average
2. The name of chemical used in preparation of ferric hydroxide sol is ....
   a. ferrous chloride c. ferric chloride
   b. iron (II) chloride d. ferric sulfate
3. In sol of ferric hydroxide the disperse phase is ....
   a. $FeCl_3$ b. $Fe(OH)_3$ c. water d. Fe
4. The type of colloid of ferric hydroxide in water is ....
   a. solid sol b. solid foam c. gel d. sol
5. Which one of the following is an example of lyophobic sol?
   a. Starch sol c. Gold sol
   b. Gelatin sol d. Gum sol
6. Which one of the following statement is INCORRECT for lyophobic sol?
   a. It has little affinity for the dispersion medium
   b. It is reversible in nature
   c. It is a hydrophobic sol
   d. It is less stable sol

Short answer question

1. What is lyophobic sol (colloid)?

Ans. A lyophobic sol is a colloidal solution in which the dispersed phase has very little or no affinity for the dispersion medium.

2. Classify the following as lyophilic sols and lyophobic sols.

Gold sol, starch sol, gum sol, ferric hydroxide sol, $Al(OH)_3$ sol.

Ans. Lyophilic sols: Starch sol, Gum sol

Lyophobic sols: Gold sol, Ferric hydroxide sol [Fe(OH) sol], Aluminium hydroxide sol [Al(OH) sol]

3. Why lyophobic sol is called irreversible sol?

Ans. Lyophobic sols are called irreversible sols because once they are precipitated or coagulated, they cannot be easily converted back into the original colloidal state by simply adding the dispersion medium.

Remark and sign of teacher: ....